

CM0246 Discrete Structures

Partial Orders

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Preliminaries

Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Introduction

We can use relations to order **some** or **all** the elements of a set.

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Example

Some order relations.

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- Schedule projects

$(a, b) \in R$ if a is a task that must be completed before the task b begins.

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Example

- $(\mathbb{Z}, \leq)$ is a poset.
- $(P(A), \subseteq)$ is a poset.

Partial Orders

Definition

Let $a, b \in \mathbb{Z}$ with $a \neq 0$. The **divisibility relation**, denoted by $|$, is defined by

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Example

- $(\mathbb{Z}^+, |)$ is a poset.
- Is $(\mathbb{N}, |)$ a poset?

Problem 6 (p. 492)

Let (A, R) be a poset. Prove that (S, R^{-1}) is also a poset, where R^{-1} is the inverse of R . The poset (S, R^{-1}) is called the dual of (S, R) .

Partial Orders

Notation

$\preceq$: Denotes an arbitrary partial order

$$a \prec b \stackrel{\text{def}}{=} a \preceq b \wedge a \neq b$$

$(A, \preceq)$: Denotes an arbitrary poset

Comparable Elements

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Let $(A, \preccurlyeq)$ be a poset. The elements $a, b \in A$ are called **comparable** iff either $a \preccurlyeq b$ or $b \preccurlyeq a$.

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Total Orders

Definition

If $(A, \preccurlyeq)$ is a poset and every two elements of A are comparable, A is called a **totally ordered set** (or **linearly ordered set**). The relation $\preccurlyeq$ is called a **total order** (or a **linear order**).

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- $(\mathbb{Z}, \leq)$ is a totally ordered set.
- $(\mathbb{Z}^+, |)$ is a not totally order set.
- Is $(P(A), \subseteq)$ a totally ordered set?

Well-Ordered Sets

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An element $a \in A$ is the **least element** (Spanish: **mínimo**) of a poset $(A, \preceq)$ iff $a \preceq b$ for all $b \in A$.

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- $(\mathbb{N}, \leq)$ is a well-ordered set.
- $(\mathbb{N}, \geq)$ is not a well-ordered set.

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- $(\mathbb{N}, \geq)$ is not a well-ordered set.
- Is $(\mathbb{Z}, \leq)$ a well-ordered set?

Hasse Diagrams

Example

Digraph for the relation $\{ (a, b) \mid a \leq b \}$ on $\{1, 2, 3, 4\}$.

See whiteboard.

Hasse Diagrams

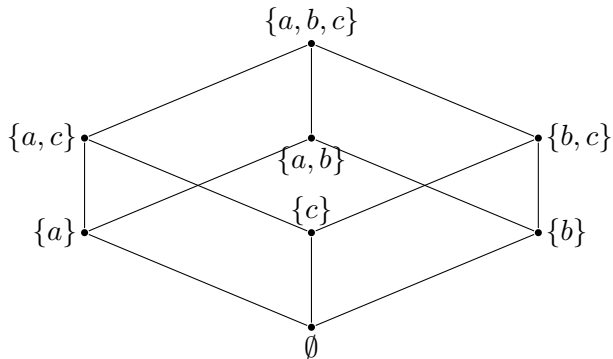
Constructing a Hasse diagram

1. Construct a digraph representation for the poset $(A, \preceq)$.
2. Remove these loops.
3. Remove all edges that must be in the partial ordering because of the presence of other edges and transitivity.
4. Arrange each edge so that its initial vertex is below its terminal vertex.
5. Remove all the arrows on the directed edges.

Hasse Diagrams

Example

Hasse diagram for the poset $(\{a, b, c\}, \subseteq)$.



Hasse Diagrams

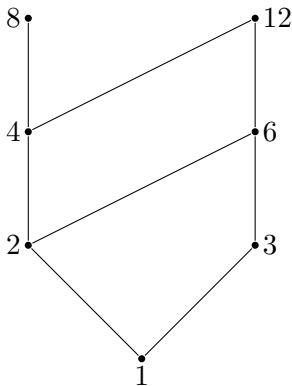
Exercise

Draw the Hasse diagram for the poset $(\{1, 2, 3, 4, 6, 8, 12\}, |)$.

Hasse Diagrams

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Draw the Hasse diagram for the poset $(\{1, 2, 3, 4, 6, 8, 12\}, |)$.



Lexicographic Ordering

Example

Let $\preccurlyeq$ be relation on $\mathbb{Z} \times \mathbb{Z}$ defined by

$$(a_1, b_1) \preccurlyeq (a_2, b_2) \stackrel{\text{def}}{=} a_1 < a_2 \text{ or } (a_1 = a_2 \text{ and } b_1 \leq b_2).$$

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- Is $(\mathbb{Z} \times \mathbb{Z}, \preceq)$ a poset?

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- Is $(3, 100) \preccurlyeq (4, 4)$?
- Is $(3, 5) \preccurlyeq (3, 4)$?
- Is $(\mathbb{Z} \times \mathbb{Z}, \preccurlyeq)$ a poset?
- Is $(\mathbb{Z} \times \mathbb{Z}, \preccurlyeq)$ a totally ordered set?

Lexicographic Ordering

Definition

Let $(A, \preceq_A)$ and $(B, \preceq_B)$ be two posets. The **lexicographic ordering** $\preceq$ on $A \times B$ is defined by:

$$(a_1, b_1) \preceq (a_2, b_2) \stackrel{\text{def}}{=} a_1 \prec_A a_2 \text{ or } (a_1 = a_2 \text{ and } b_1 \preceq_B b_2).$$

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Example

- Previous example
- Whiteboard

Lexicographic Ordering

Definition

Let $(A_1, \preccurlyeq_1), \dots, (A_n, \preccurlyeq_n)$ be n posets. The **lexicographic ordering** $\preccurlyeq$ on $A_1 \times \dots \times A_n$ is defined by:

$$(a_1, \dots, a_n) \preccurlyeq (b_1, \dots, b_n) \stackrel{\text{def}}{=} (\exists m > 0)(\forall i < m)(a_i = b_i \wedge a_m \preccurlyeq_m b_m),$$

that is, if one of the terms $a_m \preccurlyeq_m b_m$ and all the preceding terms are equal.

Lexicographic Ordering

Example

Let Σ be an alphabet defined by $\Sigma = \{0, 1\}$. The lexicographical ordering on $(\Sigma, \leq) \times (\Sigma, \leq) \times (\Sigma, \leq)$ is given by



111
110
101
100
011
010
001
000

Lexicographic Ordering

Definition

Let Σ^* be the set of all words (finite sequence of symbols) on an alphabet Σ , including the empty word denoted by λ .

A **lexicographic ordering** on Σ^* can be defined by: if the words are the same length, use the lexicographic ordering of n posets, else the shorter sequence should be padded at the end with enough "blanks" (a special symbol that is treated as smaller than every element of Σ).

Lexicographic Ordering

Example

Let Σ be an alphabet defined by $\Sigma = \{0, 1\}$. The lexicographical ordering on $\{w \in \Sigma^* \mid l(w) \leq 3\}$ is given by

1 •
|
0 •
|
ε •

111 •	• 111
110 •	• 110
11 •	• 11ε
101 •	• 101
100 •	• 100
10 •	• 10ε
1 •	• 1εε
011 •	• 011
010 •	• 010
01 •	• 01ε
001 •	• 001
000 •	• 000
00 •	• 00ε
0 •	• 0εε
λ •	• λεεε

Product Order

Definition

Let $(A, \preceq_A)$ and $(B, \preceq_B)$ be two posets. The **product order** $\preceq$ on $A \times B$ is defined by:

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Example

Whiteboard.

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Problem 33 (p. 494)

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Let $(A, \preceq_A)$ and $(B, \preceq_B)$ be two posets. We need to prove that $(A \times B, \preceq)$ is a poset, where $\preceq$ is the product order on $A \times B$.

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- Antisymmetry: If $(a_1, b_1) \preceq (a_2, b_2)$ and $(a_2, b_2) \preceq (a_1, b_1)$ then $(a_1, b_1) = (a_2, b_2)$, for all $a_1, a_2 \in A$ and $b_1, b_2 \in B$. Whiteboard.

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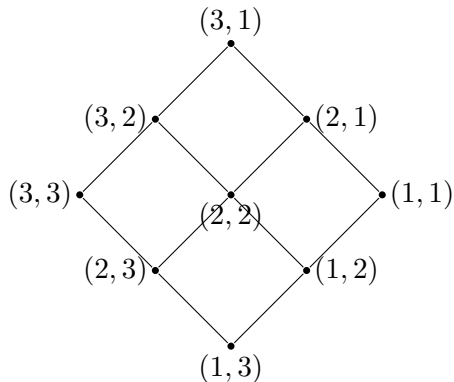
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- Transitivity: If $(a_1, b_1) \preceq (a_2, b_2)$ and $(a_2, b_2) \preceq (a_3, b_3)$ then $(a_1, b_1) \preceq (a_3, b_3)$, for all $a_1, a_2, a_3 \in A$ and $b_1, b_2, b_3 \in B$. Whiteboard. ■

Product Order

Example

Hasse diagram for the product order of the posets $(\{1, 2, 3\}, \leq)$ and $(\{1, 2, 3\}, \geq)$.



Notable Elements

Let $(A, \preceq)$ be a poset.

Definition

An element $a \in A$ is the **greatest element** (Spanish: **máximo**) of A iff $b \preceq a$ for all $b \in A$.

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An element $a \in A$ is a **maximal** of A if there is no $b \in A$ such that $a \prec b$.

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An element $a \in A$ is a **minimal** of A if there is no $b \in A$ such that $b \prec a$.

Notable Elements

Example

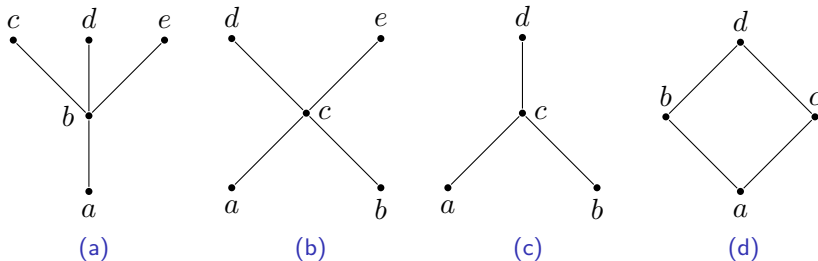


Fig.	Least element	Greatest element	Maximals	Minimals
(a)	a		c, d, e	a
(b)			d, e	a, b
(c)		d	d	a, b
(d)	a	d	d	a

Notable Elements

Let $(S, \preceq)$ be a poset and let $A \subseteq S$.

Definition

Let $u \in S$ be an element such that $a \preceq u$ for all elements $a \in A$, then u is an **upper bound** of A .

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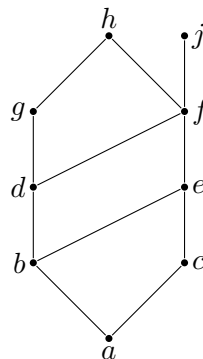
Example (using intervals of real numbers)

Whiteboard.

Notable Elements

Example

- $A = \{a, b, c\}$
Upper bounds: $\{e, f, j, h\}$
Lower bounds: $\{a\}$
- $A = \{j, h\}$
No upper bounds.
Lower bounds: $\{a, b, c, d, e, f\}$
- $A = \{a, c, d, f\}$
Upper bounds: $\{f, h, j\}$
Lower bounds: $\{a\}$



Notable Elements

Definition

An element x is the **supremum** (or the **least upper bound**) of the subset A , denoted by $\sup(A)$, iff x is an upper bound that is less than every other upper bound of A .

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Example (Using intervals of real numbers)

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Notable Elements

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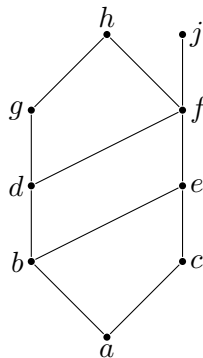
$$A = \{b, d, g\}$$

Upper bounds: $\{g, h\}$

$$\sup(A) = g$$

Lower bounds: $\{a, b\}$

$$\inf(A) = b$$

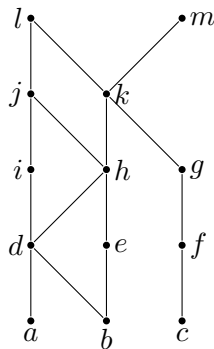


Notable Elements

Problem 26 (p. 493)

Answer these questions for the partial order represented by this Hasse diagram.

- Maximals? $\{l, m\}$
- Minimals? $\{a, b, c\}$
- Greatest element? Doesn't exist
- Least element? Doesn't exist
- Upper bounds of $\{a, b, c\}$? $\{k, l, m\}$
- $\sup(\{a, b, c\})$? k
- Lower bounds of $\{f, g, h\}$? Don't exist
- $\inf(\{f, g, h\})$? Doesn't exist

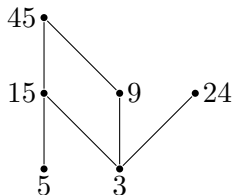


Notable Elements

Problem 27 (p. 492)

Answer these questions for the poset $(\{3, 5, 9, 15, 24, 45\}, |)$.

- Maximals? $\{24, 45\}$
- Minimals? $\{3, 5\}$
- Greatest element? Doesn't exist
- Least element? Doesn't exist
- Upper bounds of $\{3, 5\}$? $\{15, 45\}$
- $\sup(\{3, 5\})$? 15
- Lower bounds of $\{15, 45\}$? $\{3, 5, 15\}$
- $\inf(\{15, 45\})$? 15



References

Kenneth H. Rosen [1988] (2004). Matemática Discreta y sus Aplicaciones. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).